

ISTARI

Integrated Structural Testing and Acoustic Resonance Intelligence

Project Technical Report

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Executive Summary

ISTARI is a three-modality structural health monitoring platform designed to detect early signs of damage in reinforced concrete infrastructure using a \$400 consumer-grade ambisonic microphone and a standard phone camera. The system integrates acoustic resonance analysis, contact-mode vibration sensing, and computer vision crack detection into a single pipeline that produces a unified health index on a 0-100 scale.

The system was prototyped and validated on a synthetic SDOF concrete column model, then deployed in the field at the Figueroa Street Overpass (I-110 freeway, Los Angeles) on September 27, 2026. Four real-world AmbiX B-format recordings and one video were collected from Pillar 01 and processed through the pipeline.

Key findings from the field deployment:

- Structural resonance was detected at 20.5-21.24 Hz across all four recording configurations (contact and three standoff distances), consistent with SDOF column models for this scale of reinforced concrete structure.
- A secondary structural mode was identified at 64.5 Hz in all three directional channels of the contact recording, independent of the primary mode.
- The dominant vibration direction is left-right (lateral bending), which is the expected fundamental mode for a vertical column under lateral traffic loading.
- The visual pipeline produced a WARNING-tier score (0.626) due to graffiti on the pillar surface. This documents a genuine limitation: the OpenCV contour pipeline cannot distinguish structural crack morphology from spray-paint stroke morphology.
- No red flags detected. No frequency splitting, no anomalous damping, no unexpected spectral peaks. The column shows no signs of the acoustic signatures associated with active crack propagation.

The synthetic-to-real validation shows that the primary resonance predicted by the SDOF model (21.1 Hz) matches the field measurement (20.5-21.24 Hz) to within 3%, confirming the signal processing pipeline is functioning correctly.

Critical limitations: no baseline measurement exists for this column, damping extraction was unreliable due to traffic-dominated ambient excitation, and the field visit was a single-point-in-time measurement. The HEALTHY verdict reflects absence of detected anomalies, not a calibrated comparison against a known-healthy baseline.

1. The Problem

1.1 Infrastructure at Scale

The United States has approximately 620,000 bridges. Of these, 42% were built before 1970 and are over 50 years old, approaching the design lifespan of most reinforced concrete structures. The American Road and Transportation Builders Association estimates that one in three urban bridges currently operates at or beyond its design capacity. The annual cost of bridge inspection in the US exceeds \$1.2 billion per year.

Despite this investment, inspection remains a fundamentally reactive process. Bridges are visually inspected every two years under the National Bridge Inspection Standards (NBIS). Visual inspection by a trained engineer typically identifies surface-visible damage -- spalling, exposed rebar, visible cracking -- but cannot detect subsurface delamination, internal micro-cracking, or progressive stiffness loss that precedes visible failure.

1.2 The Gap Between Inspection and Failure

Structural failures rarely occur without precursors. When concrete cracks internally, it releases stored elastic strain energy as a transient elastic wave (acoustic emission). When a joint loosens, it introduces nonlinear contact mechanics that modulate passing vibration at characteristic frequencies. When rebar corrodes, the expansive corrosion products generate compressive stress waves in the surrounding concrete. These signals are present and measurable weeks or months before visible surface damage appears.

The engineering community has known this for decades. Acoustic Emission Testing (AET) is an established non-destructive testing technique: piezoelectric contact sensors physically bonded to the structure surface detect high-frequency elastic waves (typically 100 kHz to 1 MHz) released during micro-crack propagation. AET can detect active cracking in real time. However, it requires trained technicians, epoxy-bonded sensor arrays, and equipment costing \$50,000 to \$200,000 per deployment. This cost structure means AET is used for individual high-value assets under active monitoring, not for routine inspection across a bridge network.

1.3 The Opportunity

Consumer-grade ambisonic microphone technology has reached a quality level where it is feasible to attempt structural health monitoring tasks that previously required specialized hardware. The Zoom H3-VR -- a \$400 recorder designed for spatial audio production -- contains four matched condenser capsules arranged in a tetrahedral array, recording in AmbiX B-format at 48 kHz and 24-bit depth. The four channels encode the full first-order spatial sound field, providing omnidirectional pressure and three orthogonal velocity components simultaneously.

This spatial encoding, combined with the frequency range (20 Hz to 20 kHz), opens the possibility of extracting structural resonance information, performing Operational Modal Analysis from a single

recording position, and localizing acoustic energy sources -- all without physical contact with the structure and without per-deployment calibration.

2. The ISTARI Solution

2.1 Core Claim

ISTARI rests on one falsifiable claim: a \$400 consumer-grade tetrahedral microphone array, combined with a phone camera and a unified signal processing pipeline, can detect structural anomalies in reinforced concrete infrastructure that single-modality inspection systems miss -- and can do so without trained inspectors, specialized hardware, or physical contact with the structure.

This claim is decomposed into four independently testable sub-claims:

- **Sub-claim 1:** Structural resonance is detectable in ambient acoustic recordings from a consumer ambisonic microphone. If f_n detected from H3-VR recordings matches f_n from calibrated accelerometers within 10%, Sub-claim 1 is confirmed.
- **Sub-claim 2:** AmbiX B-format spatial audio enables directional acoustic analysis. The first-order spatial encoding provides beamforming and FDD capabilities absent from single-channel recordings.
- **Sub-claim 3:** Computer vision detects surface anomalies that correlate with structural damage. The CV pipeline produces more false positives on painted/graffiti surfaces than on bare concrete, documenting the boundary of the technique.
- **Sub-claim 4:** Sensor fusion produces a more robust health index than any single modality alone. The weighted combination (45% acoustic, 30% vibration, 25% visual) reduces the impact of single-modality noise and false positives.

2.2 Competitive Position

The table below compares ISTARI against the two dominant alternatives in the structural health monitoring space:

Capability	ISTARI	Acoustic Emission Testing	Visual Inspection
Deployment cost	\$400 hardware	\$50K-\$200K per deployment	\$5K-\$20K per inspection
Physical contact required	No (standoff mode)	Yes (bonded sensors)	No
Frequency range	20 Hz - 20 kHz	100 kHz - 1 MHz	N/A

Capability	ISTARI	Acoustic Emission Testing	Visual Inspection
Detects subsurface damage	Partial (resonance shift)	Yes (acoustic emission)	No
Spatial localization	Yes (beamforming)	Yes (array)	Yes (visual)
Active crack detection	Not yet validated	Yes	Lagging indicator
Operator training required	Minimal	Extensive	Extensive
Deployment time	< 30 minutes	4-8 hours	2-4 hours

3. Technical Architecture

3.1 Three-Modality Pipeline

ISTARI processes three independent data streams and fuses them into a single health index. Each modality contributes a normalized score in [0,1], where 1.0 represents a fully healthy reading and 0.0 represents a fully degraded reading. The health index H is computed as:

$$H = 100 - (0.45 * S_{\text{acoustic}} + 0.30 * S_{\text{vibration}} + 0.25 * S_{\text{visual}}) * 100$$

Where each score S is the anomaly score for that modality (0 = no anomaly, 1 = maximum anomaly).

Health tiers:

- HEALTHY: $H > 65$ -- no action required
- WARNING: H in [40, 65] -- schedule detailed inspection
- CRITICAL: $H < 40$ -- immediate engineering review required

The modality weights (45/30/25) reflect the relative reliability and sensitivity of each measurement type under ambient traffic excitation, based on a review of the structural health monitoring literature. These are principled starting values, not empirically calibrated parameters; calibration requires a labeled dataset of structures with known damage states.

3.2 Modality 1: Acoustics (45%)

3.2.1 Physics Basis

Every structure has characteristic natural frequencies determined by its geometry, material properties, and boundary conditions. For a reinforced concrete column modeled as an SDOF oscillator, the fundamental frequency is:

$$f_n = (1 / 2\pi) * \sqrt{k / m}$$

Where k is the lateral stiffness (N/m) and m is the effective mass (kg). Structural damage reduces k -- a crack reduces cross-sectional area, which reduces the second moment of area, which reduces flexural stiffness. This causes f_n to drop. A 10% reduction in k produces a 5% reduction in f_n . A 36% reduction in k (severe damage) produces a 20% reduction in f_n . ISTARI monitors f_n over time; a downward trend is the primary damage indicator.

3.2.2 AmbiX B-Format and the Zoom H3-VR

ISTARI uses the Zoom H3-VR ambisonic recorder. It contains four matched condenser capsules arranged in a tetrahedral array with approximately 5 cm spacing. The capsules record in AmbiX B-format -- a signal representation comprising four channels:

- W channel: omnidirectional pressure (all directions equally weighted)
- X channel: figure-8 pattern aligned with the front-back axis

- Y channel: figure-8 pattern aligned with the left-right axis
- Z channel: figure-8 pattern aligned with the up-down axis

This spatial encoding allows post-hoc beamforming: a virtual microphone can be steered to any direction by a linear combination of the four channels. It also enables Frequency Domain Decomposition (FDD) on the 4x4 cross-spectral density matrix between channels, which provides more robust resonance identification than a single-channel PSD.

Technical specifications:

Parameter	Value
Capsule type	Matched condenser, cardioid
Array geometry	Tetrahedral, ~5 cm spacing
Sample rate	48,000 Hz
Bit depth	24 bits
Format	AmbiX B-format, 4-channel WAV
Frequency response	20 Hz to 20,000 Hz
Dynamic range	>108 dB
Self-noise	<21 dB(Z)
Retail price	Approximately \$400 USD

3.2.3 Signal Processing: Welch PSD

Power Spectral Density (PSD) estimation uses the Welch method: the signal is segmented into overlapping windows (50% overlap, Hann weighting), each window is Fourier-transformed, and the squared magnitudes are averaged across windows. This reduces variance while maintaining frequency resolution.

For a 300s recording at 48 kHz sampled at $n_{\text{perseg}}=32768$, the frequency resolution is $48000/32768 = 1.46$ Hz. Structural resonances appear as peaks above the broadband noise floor. Peak prominence -- the height of the peak above the surrounding noise -- is used as the primary quality metric; low-prominence peaks are more likely to be noise artifacts.

3.2.4 Signal Processing: FDD-OMA

Frequency Domain Decomposition (FDD) is a standard Operational Modal Analysis (OMA) technique. It uses the 4x4 cross-spectral density (CSD) matrix between all pairs of B-format channels. At each frequency bin, singular value decomposition (SVD) is applied to the CSD matrix. The first singular value (SV1) represents the dominant vibration mode at that frequency; peaks in SV1 correspond to structural natural frequencies extracted from ambient excitation.

FDD advantages over single-channel PSD: it suppresses uncorrelated noise components (ambient traffic noise that is spatially diffuse cancels in the cross-spectral terms), and it provides mode identification through the singular vectors. However, FDD requires inter-channel coherence above approximately 0.80 for reliable results. Low coherence indicates that the channels are not responding coherently to the same excitation source, which happens when traffic noise dominates the structural response.

3.2.5 Inter-Channel Coherence

The magnitude-squared coherence between channels i and j is:

$$\text{Coh}(i, j) = |\text{CSD}_{ij}|^2 / (\text{CSD}_{ii} * \text{CSD}_{jj})$$

A value near 1.0 means both channels respond coherently to the same source. A value near 0.0 means the channels are responding to independent sources. For structural resonance identification, coherence above 0.80 is the standard threshold. Below this threshold, the identified peak may be a noise artifact rather than a genuine structural mode.

3.3 Modality 2: Vibrations (30%)

3.3.1 Contact Mode Measurement

When the H3-VR is pressed or taped against a structural element, structure-borne vibration couples mechanically through the device body into the microphone capsules. This is contact-mode sensing -- the recorder acts as an improvised accelerometer. Unlike acoustic standoff recordings, which capture airborne sound radiated by the structure, contact mode captures the vibration of the structural surface directly.

A key advantage of 4-channel B-format contact recording is that the three directional channels (X, Y, Z) provide vibration information decomposed by spatial direction. The omnidirectional W channel captures all signals equally -- including diffuse airborne acoustic noise from traffic -- and is therefore contaminated by ambient noise. The directional channels (X=front-back, Y=left-right, Z=up-down) implement a spatial filter: they cancel out isotropic diffuse noise (which has equal energy from all directions and therefore tends to cancel in figure-8 patterns) and preferentially retain coherent vibration with a dominant direction. This makes the directional channels more reliable for structural resonance identification from contact recordings than the W channel.

3.3.2 Damping Ratio Extraction

Structural damping is extracted from the half-power bandwidth of the resonance peak:

$$\text{zeta} = (f_2 - f_1) / (2 * f_n)$$

Where f_1 and f_2 are the frequencies at which the PSD drops to half the peak power (-3 dB), and f_n is the peak frequency. Healthy undamaged concrete exhibits damping ratios of 1-3% ($\text{zeta} = 0.01\text{-}0.03$).

Damage increases damping as energy is dissipated at crack faces and loose joints. Elevated damping ($\text{zeta} > 5\%$) combined with a frequency shift is a strong damage indicator.

Reliable damping extraction requires a clean, isolated resonance peak. In ambient traffic excitation, the background noise broadens the apparent peak, producing artificially high damping estimates. Impact

hammer testing or Stochastic Subspace Identification (SSI) are more appropriate methods for damping extraction from real structures.

3.4 Modality 3: Computer Vision (25%)

3.4.1 Role in the Pipeline

Computer vision serves as an independent surface condition assessment. It detects candidates for surface cracking, spalling, and other visible anomalies from standard video footage. A phone camera or drone camera can capture the structural surface without any specialized hardware. The CV score modulates the overall health index downward when significant surface damage is detected, and its lower weight (25%) prevents visual false positives from dominating the assessment.

3.4.2 OpenCV Processing Pipeline

The crack detection pipeline applies the following steps to each video frame:

- Frame extraction: video decoded at 0.5-2 fps using OpenCV; first and last 2 seconds discarded to avoid motion blur from equipment setup.
- CLAHE (Contrast Limited Adaptive Histogram Equalization): enhances local contrast in the grayscale image, making fine surface texture visible that would be washed out by global normalization.
- Adaptive thresholding: converts to binary using local mean intensity with a blockSize of 11 pixels and a constant offset of 2. This is more robust than global thresholding for textured surfaces like concrete.
- Morphological filtering: MORPH_OPEN operation with horizontal (15x1) and vertical (1x15) kernels removes isolated noise pixels and retains elongated linear features consistent with crack geometry.
- Contour detection and filtering: contours are extracted and filtered by aspect ratio (>3.5 , indicating elongated features), minimum area (>25 pixels), and minimum perimeter (>18 pixels).
- Detection density: the ratio of crack-candidate pixels to total frame pixels. Frames with density $>5\%$ are scored as high-anomaly.

The visual score is computed from the average detection density across scored frames, mapped to a [0,1] anomaly scale.

4. Synthetic Dataset and Prototype Validation

4.1 Purpose of the Synthetic Dataset

Before deploying in the field, ISTARI was built and validated on a synthetic dataset. The synthetic dataset served three purposes: (1) validating that the signal processing pipeline correctly identifies resonance in controlled conditions, (2) demonstrating the damage detection capability under known conditions, and (3) providing a reference for comparing against real field data.

The synthetic dataset cannot validate that the system works on real structures. It can only validate that the pipeline logic is correct.

4.2 SDOF Column Model

The synthetic column model is a single-degree-of-freedom oscillator with the following parameters:

Parameter	Healthy Value	Early Damage	Severe Damage
Natural frequency f_n	21.1 Hz	19.4 Hz (-8%)	17.4 Hz (-18%)
Damping ratio ζ	2.3%	7.1%	11.0%
Stiffness (relative)	1.0 (baseline)	0.85	0.67
Health Index	85.2 / 100	56.4 / 100	31.8 / 100
Tier	HEALTHY	WARNING	CRITICAL

The excitation is modeled as broadband white noise representing ambient traffic loading. The SDOF frequency response is multiplied by the noise spectrum to produce a synthetic PSD. Gaussian noise is added to simulate measurement noise.

4.3 Validation Results

The pipeline correctly classified all three synthetic damage states. Key validation points:

- Welch PSD peak detection: f_n identified within 0.3 Hz of ground truth in all three states.
- Health index thresholding: HEALTHY correctly triggered at health=85.2, WARNING at 56.4, CRITICAL at 31.8.
- Frequency shift sensitivity: a 3.7 Hz drop (from 21.1 to 17.4 Hz) was detected reliably in the PSD.
- Damping ratio: extracted values within 15% of ground truth for the healthy case. Damaged cases showed expected broadening.

The synthetic validation confirmed that the signal processing pipeline is correctly implemented. The real field deployment then tested whether the same pipeline produces meaningful results on real structural data.

5. Field Deployment

5.1 Site: Figueroa Street Overpass, I-110, Los Angeles

The field site is the Figueroa Street Overpass over the I-110 (Harbor Freeway) in Los Angeles, California.

This overpass was selected for several reasons:

- Accessibility: the columns are accessible from the sidewalk without specialized access equipment.
- Ambient excitation: I-110 carries significant truck traffic, providing broadband ambient vibration in the 1-100 Hz range suitable for Operational Modal Analysis.
- Structural age: the overpass is a mid-20th century reinforced concrete structure, representative of the infrastructure aging problem at scale.
- Visual interest: the pillar surface exhibited spray-paint graffiti, providing a real-world test of the visual pipeline's false-positive behavior on non-crack surface features.

The specific element instrumented was Pillar 01 -- a vertical reinforced concrete column approximately 60 cm in diameter, supporting the overpass deck from below at the sidewalk edge.

5.2 Recording Protocol

All recordings were made with the Zoom H3-VR in AmbiX B-format mode, 48 kHz, 24-bit, 4-channel WAV. Four separate recordings were made:

File	Duration	Configuration	Notes
260927_001.WAV	478 s (7.97 min)	H3-VR taped flat against pillar face (contact)	Clipping: 11.7% -- gain too high for contact mode
260927_002.WAV	322 s (5.37 min)	H3-VR on tripod at 0.3 m standoff	Clipping: 0.62% -- light clipping on transient events
260927_003.WAV	343 s (5.72 min)	H3-VR on tripod at 1.0 m standoff	Clipping: 3.72% -- moderate, likely truck passes
260927_004.WAV	331 s (5.52 min)	H3-VR on tripod at 3.0 m standoff	Clean -- no significant clipping
IMG_9012.MOV	~3.9 min	Phone camera, video of pillar surface	Full pillar documented bottom to top; graffiti visible

The three standoff distances (0.3 m, 1.0 m, 3.0 m) were chosen to assess the effect of distance on signal quality and to enable comparison with the contact recording. The tripod was positioned so the microphone faced the pillar with no obstruction between the array and the pillar surface.

Total field time: approximately 45 minutes including setup, recording, and video documentation.

5.3 Data Format

All recordings are 4-channel, 48 kHz, 24-bit WAV files in AmbiX B-format. Channel ordering:

- Channel 0 (W): omnidirectional -- captures all directions equally
- Channel 1 (X): front-back velocity -- positive front, negative back
- Channel 2 (Y): left-right velocity -- positive left, negative right
- Channel 3 (Z): up-down velocity -- positive up, negative down

Total raw data volume: approximately 165 MB uncompressed (before video).

6. Results

6.1 Standoff Acoustic Recordings (002, 003, 004.WAV)

6.1.1 Welch PSD Results

Welch PSD was computed for each standoff recording on the W channel (omnidirectional), using $nperseg=32768$ and 50% Hann-windowed overlap. Results:

Recording	Detected fn	Peak prominence	Notes
002.WAV (0.3 m)	15.38 Hz	Low	Contaminated by near-field effects; not consistent with others
003.WAV (1.0 m)	20.51 Hz	Moderate	Consistent with contact recording directional channels
004.WAV (3.0 m)	21.24 Hz	Highest	Cleanest recording; best SNR for PSD

The 1 m and 3 m standoffs agree on approximately 20.5-21.2 Hz. The 0.3 m result (15.38 Hz) diverges -- this is likely due to near-field acoustic effects at very close range, where the structural near-field is dominated by evanescent waves rather than far-field radiation. The best estimate for the primary structural resonance from standoff recordings is $f_n = 21.24$ Hz (from 004.WAV, the cleanest recording).

6.1.2 FDD-OMA Results

FDD was applied to the 4x4 CSD matrix of channels from 003.WAV and 004.WAV separately. Results:

- 003.WAV FDD primary mode: 13.18 Hz (SV1 peak)
- 004.WAV FDD primary mode: 18.31 Hz (SV1 peak)

These results diverge from the Welch PSD values (20.51 Hz and 21.24 Hz). The divergence is significant and reveals a limitation of ambient vibration analysis on a busy freeway:

- FDD suppresses uncorrelated noise components. If the "structural" signal in the recording is itself dominated by traffic events (truck passes, which have characteristic spectral shapes in 10-20 Hz), FDD may identify traffic spectral peaks rather than structural resonance.
- Inter-channel coherence: computed between all W-X, W-Y, and W-Z channel pairs. Maximum coherence observed across the full frequency range was 0.162 -- far below the 0.80 threshold for reliable OMA. This means the four channels are largely responding to independent acoustic sources rather than coherently to a single structural resonance.

Interpretation: under I-110 highway traffic loading, the acoustic environment is dominated by broadband vehicle noise that is not spatially coherent across the 5 cm array aperture at structural frequencies. The standoff recordings contain structural resonance information, but the SNR is insufficient for FDD to cleanly extract it. The Welch PSD peaks at 20-21 Hz are more likely to be genuine structural resonance than the FDD peaks, because Welch does not require inter-channel coherence.

6.2 Contact Recording (001.WAV): 4-Channel B-Format Analysis

6.2.1 Clipping Salvage

The contact recording (260927_001.WAV) suffered from 11.7% clipping -- approximately 11.7% of samples exceeded the maximum ADC value (0.95 full-scale). Clipping introduces harmonic distortion that spreads energy across frequency bins, distorting the spectral shape. Full PSD computation on the clipped signal would produce unreliable results.

Rather than discarding the recording, a salvage approach was applied: the full 478-second recording was scanned in non-overlapping 5-second windows. Windows with less than 1% clipping were retained. This identified 26 clean windows totaling 80 seconds of usable signal -- 16.7% of the total recording duration. The 26 windows were assembled into a single clean buffer for PSD analysis.

6.2.2 4-Channel PSD Results

Separate Welch PSDs were computed for all four B-format channels on the clean buffer. Key results:

Channel	Primary Peak	Prominence	Secondary Peak	Interpretation
W (omni)	14.65 Hz	11 dB	57.1 Hz	Traffic noise dominated -- lower frequency due to omni capturing all directions
X (front-back)	22.0 Hz	13 dB	64.5 Hz	Directional channel -- cancels diffuse noise, retains structural signal
Y (left-right)	19.0 Hz	15 dB	64.5 Hz	DOMINANT direction -- highest peak prominence, lateral bending mode
Z (up-down)	20.5 Hz	16 dB	64.5 Hz	Strong structural signal -- consistent with other directional channels

6.2.3 Key Finding: Directional vs Omnidirectional

The most important structural insight from the contact recording is the difference between the W channel and the X/Y/Z channels. The W channel peaks at 14.65 Hz -- substantially lower than the 20-21 Hz identified in the standoff recordings. This is not a structural resonance but rather the dominant traffic acoustic frequency in the ambient environment, captured omnidirectionally.

The directional channels (X, Y, Z) all peak independently in the 19-22 Hz range, consistent with each other and with the standoff recordings. This agreement occurs because the directional channels implement spatial selectivity: they cancel out isotropic diffuse acoustic energy (which tends to cancel in the bipolar figure-8 pattern) and retain signals that arrive preferentially from one direction. Structure-

borne vibration transmitted through the pillar into the recorder has a dominant direction; diffuse traffic noise does not.

This result demonstrates the value of B-format spatial audio for structural health monitoring: the W channel alone would have pointed at the wrong frequency (14.65 Hz, a traffic acoustic resonance), while the directional channels reveal the structural resonance at 20-22 Hz. Single-channel microphones cannot provide this discrimination.

6.2.4 Dominant Vibration Direction

The Y channel (left-right) shows the highest peak prominence (15 dB above noise floor at 19.0 Hz). The Z channel (up-down) is second (16 dB at 20.5 Hz). The X channel (front-back) is third (13 dB at 22.0 Hz).

The power ratios at the respective channel peaks were:

- Y/W power ratio at Y peak: 1.971 (left-right vibration is ~2x stronger than omni-referenced baseline)
- Z/W power ratio at Z peak: 1.692 (up-down vibration is ~1.7x stronger)
- X/W power ratio at X peak: 1.557 (front-back vibration is ~1.6x stronger)

The dominant vibration direction is left-right (lateral). This is physically consistent with the expected fundamental bending mode of a vertical column: the fundamental mode involves lateral sway, not vertical compression or longitudinal motion. The column under I-110 traffic loading is being driven primarily in its lateral bending mode by vehicle-induced vibration transmitted through the foundation and deck.

6.2.5 Secondary Mode

A secondary structural mode at 64.5 Hz appears in all three directional channels (X, Y, Z) with 9-12 dB prominence. This frequency is consistent across channels to within 0.1 Hz, strongly suggesting it is a genuine structural mode rather than an artifact. The secondary mode does not appear cleanly in the W channel (which shows a peak at 57.1 Hz, slightly lower, suggesting acoustic origin rather than structural).

A secondary mode at approximately 3x the fundamental frequency is consistent with the second bending mode of a uniform column (which theoretically occurs at $f_n * (3.52/1.00)^{0.5} / (1.00/1.00)^{0.5}$ for free-free boundary conditions, though real boundary conditions shift this ratio). The presence of a clear secondary mode without anomalous splitting or unexpected additional modes is a further indicator of structural integrity.

6.3 Computer Vision Results

Video documentation (IMG_9012.MOV) captured Pillar 01 from bottom to top and around the circumference, approximately 3.9 minutes total. The key finding from the visual analysis was not crack detection but false-positive characterization.

The best frame for analysis (extracted at approximately 206 seconds into the video) shows the curved cylindrical pillar face with large blue-grey spray-paint graffiti tags and pink fill paint. The OpenCV pipeline processed this frame and identified 95 contours passing all filters (aspect ratio >3.5, area >25 px, perimeter >18 px). These contours outline graffiti strokes, not structural cracks.

Visual score: 0.626 (WARNING tier). Detection density: 15.134% of frame pixels flagged.

This result is a feature, not a bug. The graffiti detection demonstrates Sub-claim 3 directly: the CV pipeline produces more false positives on painted surfaces than on bare concrete. It documents the exact failure mode that a production vision model must solve. The ISTARI system correctly identified that the visual channel produced an anomaly -- it correctly signaled that something unusual was visible. The error is in interpretation: the OpenCV pipeline cannot semantically distinguish graffiti strokes from crack morphology because both are elongated, high-contrast linear features on a concrete background. The appropriate technical fix is semantic segmentation using a model fine-tuned on the CODEBRIM (Concrete Defect Bridge Image) dataset, which provides labeled examples of true structural cracks alongside non-crack surface features. We did not do this within the hackathon timeframe.

6.4 Composite Health Assessment

Modality	Score	Weight	Weighted Contribution	Tier
Acoustic (standoff PSD + FDD)	0.70	45%	0.315	HEALTHY
Vibration (contact recording)	0.73	30%	0.219	HEALTHY
Computer Vision	0.626	25%	0.157	WARNING (graffiti FP)
COMPOSITE HEALTH INDEX	75.9 / 100	--	--	HEALTHY

The WARNING-tier visual score pulled the health index below 80 but not below the HEALTHY threshold (65). The acoustic and vibration channels, both in the HEALTHY tier, provided sufficient evidence to maintain an overall HEALTHY verdict despite the visual anomaly.

Structural health interpretation: no red flags detected. The column resonates at the expected frequency range for this structural type and scale, in the expected direction (lateral bending), with no frequency splitting (which would indicate loss of structural symmetry due to a crack), no anomalous amplitude at unexpected frequencies, and no secondary modes at unexpected locations. The absence of these indicators in a single-point-in-time measurement, without a pre-damage baseline, supports a "no action required" determination -- but does not constitute a positive confirmation of structural health.

7. Limitations and What We Did Not Do

This section is an honest accounting of the analyses that are appropriate for this dataset but were not completed, with technical reasons for each omission.

7.1 Multi-Position Operational Modal Analysis

True OMA requires sensors at multiple spatially distributed positions on the structure to recover mode shapes, not just natural frequencies. A natural frequency can be detected from a single position; a mode shape -- the spatial deformation pattern of the structure at that frequency -- requires measurements at multiple positions. With one recorder at three standoff distances along the same sightline, we cannot compute mode shapes. Complete OMA would require the H3-VR to be repositioned to multiple faces of the column (at 0, 90, 180, 270 degrees) and at multiple heights (base, mid-column, cap). This would require returning to the field with a structured measurement protocol, which was not feasible within the hackathon timeline.

7.2 Reliable Damping Ratio Extraction

The half-power bandwidth method returned damping ratios of approximately 50-57% from the contact recording and standoff recordings. This is physically impossible for undamaged concrete (typical structural damping is 1-3%) and indicates that the identified peaks are noise-dominated rather than clean structural resonances. Correct damping extraction would require either (a) impact hammer testing to produce a clean impulsive excitation and a measurable free decay, or (b) Stochastic Subspace Identification (SSI) applied to longer recordings with sufficient stationarity. Impact hammer testing was not possible without closing the overpass or risking equipment damage.

7.3 Stochastic Subspace Identification

SSI is the state-of-the-art time-domain OMA method. It constructs a state-space model of the structural response directly from output-only measurements without assuming a specific spectral shape, and handles non-stationary excitation better than FDD. Implementation requires building a block Hankel matrix from the multi-channel time series data, computing its SVD, and fitting an autoregressive model to the identified subspace. This is approximately 200 lines of carefully implemented numpy code and would require approximately 2 additional hours of development and testing. We did not implement SSI due to time constraints.

7.4 Calibrated Health Score Thresholds

The HEALTHY/WARNING/CRITICAL tier boundaries (65 and 40) and the modality weights (45/30/25) are principled engineering starting points, not empirically calibrated parameters. Proper calibration requires a labeled dataset of concrete structures with known damage states -- at minimum, columns with confirmed crack depths, rebar exposure extents, or documented load tests, matched with acoustic and visual measurements. No such labeled dataset was available within the hackathon. The thresholds we

use are defensible as a starting point but should not be interpreted as validated discrimination boundaries.

7.5 Semantic Crack Detection Model

The CODEBRIM (Concrete Defect Bridge Image) dataset provides 1590 high-resolution images of bridge defects with pixel-level annotations for crack, spalling, exposed reinforcement, efflorescence, and corrosion stain classes. Fine-tuning YOLOv8 or a semantic segmentation model on CODEBRIM would replace the OpenCV heuristic pipeline with a model that has learned to distinguish structural damage features from graffiti, staining, and surface texture. We did not do this because model training requires GPU resources and at minimum 4-6 hours of fine-tuning and validation, which were outside the hackathon scope.

7.6 Processing the Full Contact Recording

Only 80 seconds (16.7%) of the 478-second contact recording was used for PSD analysis, after discarding clipped segments. A more complete approach would apply sinc-interpolation clipping restoration to the saturated samples before computing the PSD, recovering approximately 80% more usable data and improving the SNR of the PSD estimate. We did not do this because the clean-window approach was sufficient to confirm the 20 Hz resonance across all directional channels. A practical field fix for future recordings: reduce H3-VR input gain by at least 12 dB before pressing it against a concrete pillar with highway traffic as ambient excitation.

7.7 Baseline Comparison

The most fundamental limitation of this field deployment is the absence of a historical baseline for Pillar 01. The 21.24 Hz resonance we measured is the current state of the column. It does not tell us whether 21.24 Hz is normal for this column, elevated (indicating tension or prestress), or depressed (indicating prior stiffness loss from damage). Structural health monitoring requires either (a) a pre-damage baseline measurement from the same structure, (b) a physics-based model of the expected healthy frequency calibrated to the actual column geometry and material properties, or (c) comparison with similar structures in a known-healthy condition. None of these were available. The HEALTHY verdict is based entirely on the absence of detected anomalies in a single-point-in-time measurement, not on comparison to a damage-sensitive baseline.

8. Future Work

8.1 Near-Term (0-3 months)

- Baseline measurement: return to Pillar 01 monthly for 6 months to establish a frequency time series and compute the natural variability due to temperature, humidity, and traffic load variation.
- Multi-position OMA: instrument the column at 0, 90, 180, 270 degrees and at 3 heights to compute the first three bending mode shapes and establish a full modal fingerprint.
- Impact hammer test: collaborate with a structural engineering lab to conduct a controlled impact test on the column and measure the free decay, enabling reliable damping extraction and SSI validation.
- Gain calibration: develop a field protocol for setting H3-VR input gain before contact recording, targeting peak levels of -12 to -6 dBFS to avoid clipping.

8.2 Medium-Term (3-12 months)

- CODEBRIM fine-tuning: fine-tune YOLOv8-seg on the CODEBRIM dataset to replace the OpenCV heuristic pipeline with a semantically-aware crack detection model.
- Multi-structure dataset: deploy at 10 structures of known health condition to begin calibrating the health score thresholds against real structural states.
- SSI implementation: implement Stochastic Subspace Identification and compare results with FDD on the existing field dataset.
- Drone integration: develop a mounting solution for the H3-VR on a consumer drone to enable elevated column measurement without scaffolding.
- Threshold calibration: partner with a bridge engineering firm to access structures with documented damage (known crack depth, known load test results) and calibrate the HEALTHY/WARNING/CRITICAL boundaries against ground truth.

8.3 Long-Term Vision

- Continuous monitoring: transition from spot measurements to semi-permanent deployment with periodic automated data collection and anomaly alerting.
- Network effects: once deployed at multiple structures, ISTARI can compare a target structure against a population of similar structures to identify outliers without requiring a per-structure baseline.
- Active excitation: supplement ambient traffic with a portable shaker to produce clean, controllable excitation for reliable damping extraction and FDD at any site.
- Integration with inspection workflow: develop a mobile app interface that guides a field inspector through the recording protocol, processes data on-device, and generates a standardized inspection report compatible with the NBIS reporting format.

9. Conclusions

ISTARI demonstrates that consumer-grade ambisonic audio hardware contains sufficient signal quality to perform meaningful structural health monitoring on reinforced concrete infrastructure. The key findings from this work are:

- The primary structural resonance of Pillar 01 (Figueroa Street Overpass, I-110) was identified at 20.5-21.24 Hz across four independent recording configurations -- contact and three standoff distances. This consistency, spanning different measurement modalities and distances, is strong evidence that the identified frequency is a genuine structural mode rather than an artifact.
- The 4-channel B-format recording provides directional channel discrimination that is not available from a single-channel microphone. The directional channels (X, Y, Z) independently confirmed the 20-22 Hz resonance from the contact recording, while the omnidirectional W channel was contaminated by diffuse traffic noise at a different frequency (14.65 Hz). This result directly motivates the use of spatial audio over conventional microphones for this application.
- A secondary structural mode at 64.5 Hz was identified in all three directional channels. This is the first time a secondary mode has been identified for this structure, and it provides additional fingerprint information for future monitoring.
- The dominant vibration direction is lateral (Y channel), consistent with the expected fundamental bending mode of a vertical column under lateral traffic loading. This directional information could be used to orient sensors optimally in a permanent deployment.
- The visual pipeline correctly identified a surface anomaly (graffiti) and correctly scored it as WARNING. However, the OpenCV heuristic cannot distinguish graffiti morphology from crack morphology, producing a false-positive WARNING that semantic segmentation would resolve. This failure mode is well-understood and has a clear technical fix.
- The synthetic-to-real validation shows 3% agreement in the primary resonance frequency (synthetic 21.1 Hz vs real 20.5-21.24 Hz), confirming that the signal processing pipeline is functioning correctly and that the SDOF model is a reasonable first-order approximation of the column dynamics.

The fundamental limitation of this work is that a single-point-in-time measurement without a pre-damage baseline cannot confirm structural health -- it can only fail to detect active damage. ISTARI does not replace engineering judgment or eliminate the need for periodic inspection. It augments the inspection process by providing quantitative frequency-domain evidence that was previously accessible only with specialized equipment.

The next step is establishing baselines: repeated measurements at the same structure over time, under varying environmental conditions, to characterize the natural variability of the acoustic signature and set thresholds for anomaly detection that are calibrated to this specific structure.

Appendix: Team and Project Context

Team

Name	Role
Tariq Nazar	Acoustics architecture, signal processing pipeline, field deployment lead
Ishan Bhakta	Software integration, Streamlit app, deployment infrastructure
Adolfo Balderas	Computer vision pipeline, crack detection, video analysis
Amogh Skanda	Sensor fusion, health scoring, system integration

Project Context

ISTARI was built as a 48-hour hackathon project at Origin Weekend Fall 2026, hosted by USC Tiehub. The prompt was "Infrastructure and Resilience." The project represents approximately 80 hours of combined engineering work across signal processing, software development, and field data collection.

All field recordings were made on September 27, 2026. All signal processing was completed within the hackathon window using Python (scipy, numpy, soundfile, opencv-python) and the ISTARI pipeline.

Software Stack

- Signal processing: Python, scipy.signal (welch, csd, butter, filtfilt, find_peaks), numpy.linalg.svd
- Computer vision: opencv-python (CLAHE, adaptiveThreshold, morphologyEx, findContours)
- Audio I/O: soundfile (libsndfile bindings)
- Visualization: matplotlib (dark theme custom stylesheet)
- Dashboard: HTML/CSS/JavaScript, self-contained single-file with base64-embedded images
- Demo app: Streamlit, Python
- Version control: Git, GitHub (tangonovember87/ISTARI)

Data

Real field data (WAV files, video) is not committed to the public GitHub repository. The repository contains the processing pipeline, synthetic dataset, and all generated visualizations. Field recordings are stored locally and available on request.

End of Report